

Supplementary Table S3. RNA-Seq Quality Control and Alignment Statistics for All 10 Samples

| Sample ID   | Condition | Total Reads | Uniquely Mapped | Uniquely Mapped (%) | Multi-mapped | Multi-mapped (%) | Unmapped  | Unmapped (%) | Overall Alignment Rate (%) |
|-------------|-----------|-------------|-----------------|---------------------|--------------|------------------|-----------|--------------|----------------------------|
| SRR31454336 | Control   | 23,414,870  | 20,667,185      | 88.27%              | 713,518      | 3.05%            | 2,034,167 | 8.69%        | 91.31%                     |
| SRR31454337 | Control   | 31,269,492  | 23,053,792      | 73.73%              | 1,512,904    | 4.84%            | 6,702,796 | 21.44%       | 78.57%                     |
| SRR31454338 | Control   | 23,429,101  | 21,273,120      | 90.80%              | 560,822      | 2.39%            | 1,595,159 | 6.81%        | 93.19%                     |
| SRR31454339 | Control   | 24,507,905  | 21,848,952      | 89.15%              | 517,575      | 2.11%            | 2,141,378 | 8.74%        | 91.26%                     |
| SRR31454340 | Control   | 21,664,508  | 19,605,883      | 90.50%              | 474,472      | 2.19%            | 1,584,153 | 7.31%        | 92.69%                     |
| SRR31454341 | Infected  | 22,203,024  | 19,770,636      | 89.04%              | 448,277      | 2.02%            | 1,984,111 | 8.94%        | 91.06%                     |
| SRR31454342 | Infected  | 21,019,275  | 18,882,155      | 89.83%              | 437,740      | 2.08%            | 1,699,380 | 8.08%        | 91.92%                     |
| SRR31454343 | Infected  | 38,375,981  | 34,889,395      | 90.91%              | 1,501,724    | 3.19%            | 1,984,862 | 5.17%        | 94.10%                     |
| SRR31454344 | Infected  | 20,713,850  | 18,799,132      | 90.76%              | 708,398      | 3.42%            | 1,206,320 | 5.82%        | 94.18%                     |
| SRR31454345 | Infected  | 21,217,322  | 19,346,056      | 91.18%              | 769,417      | 3.63%            | 1,101,849 | 5.19%        | 94.81%                     |

Note: Alignment was performed using HISAT2 (v2.2.1) against the Gallus gallus reference genome (GalGal6, GCF\_016699485.2). Blue shading = Control samples; Orange shading = APEC-infected samples. SRR31454337 showed a lower alignment rate (78.57%) compared to other samples, which may reflect higher rRNA contamination or sequencing batch effects; this sample was retained in the analysis as it passed DESeq2 outlier detection.